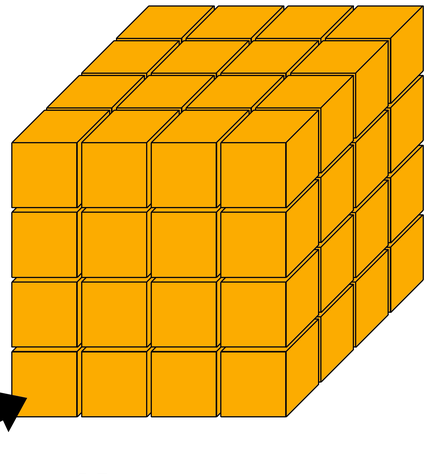
① Find a polynomial approximant on every subcube



$$f|_{I_j^N} \approx p_j^N$$

Producing the global polynomial

$$p^N = \sum_{j \in \{1,\dots,N\}^d} p_j^N \chi_j$$

such that

$\left\| f - p^N \right\|_{W^{k,\infty}([0,1]^d)}$ is small.

② Approximate the partitions of unity using tanh functions

$$p^N = \sum_{j \in \{1,\dots,N\}^d} p_j^N \chi_j$$

$$\tilde{p}^N = \sum_{j \in \{1,\dots,N\}^d} p_j^N \Phi_j^{N,d}$$

Making

$\left\| p^N - \tilde{p}^N \right\|_{W^{k,\infty}([0,1]^d)}$ small.

③ Find the EML tree

satisfying exactly

$$\tilde{p}^N = T^N,$$

Put EML tree sketch.

making

$$\left\| \tilde{p}^N - T^N \right\|_{W^{k,\infty}([0,1]^d)} = 0.$$

Aim: $$\left\| f - T^N \right\|_{W^{k,\infty}([0,1]^d)} \leq \overbrace{\left\| f - p^N \right\|_{W^{k,\infty}([0,1]^d)}}^{①} + \overbrace{\left\| p^N - \tilde{p}^N \right\|_{W^{k,\infty}([0,1]^d)}}^{②} + \overbrace{\left\| \tilde{p}^N - T^N \right\|_{W^{k,\infty}([0,1]^d)}}^{③}$$